

Course: B. Tech.

Semester : IV

Branch : Electrical and Allied Engineering

Subject Code & Name: Network Theory (BTEEC401)

Max Marks: 60

Date: 16/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

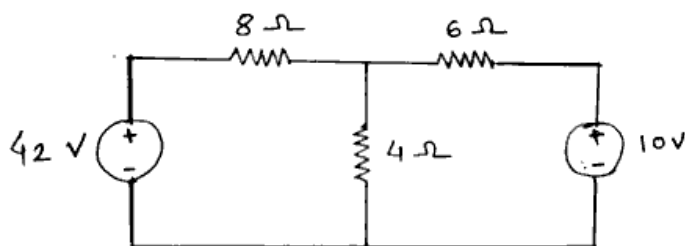
12

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|---|---------------|---|
| A) Explain the types of dependent and independent sources. | Understanding | 6 |
| B) Three resistance are connected in delta obtain the star equivalent of the network. | Remembering | 6 |
| C) State and Explain classification of Elements with their examples. | Understanding | 6 |

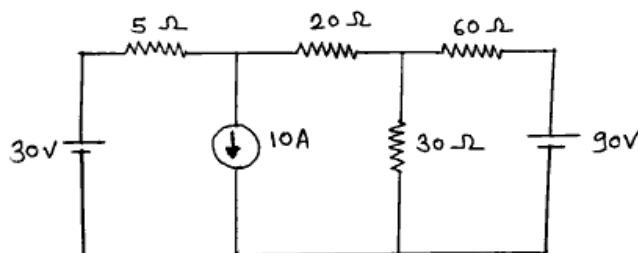
Q.2 Solve Any Two of the following.

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| A) Using the mesh analysis, find the current in various branches in the circuit. | Remembering | 6 |
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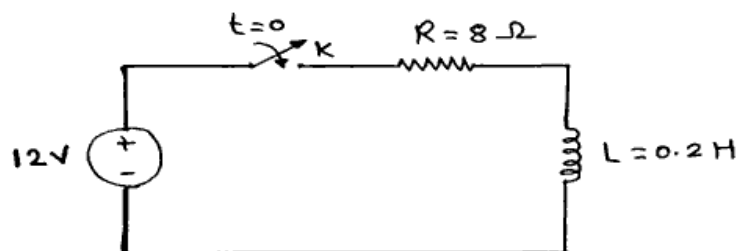


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| B) State and Explain Maximum Power Transfer Theorem. | Understanding | 6 |
| C) Find the value of current flowing through the 30 Ω resistor by using super mesh analysis. | Remembering | 6 |

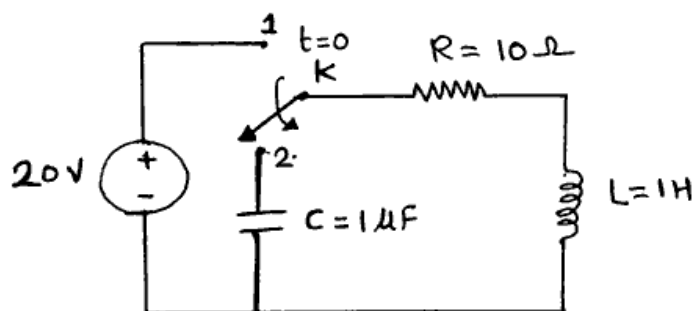


Q. 3 Solve Any Two of the following. 12

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| A) Obtain transient response of driven R-L series circuit. | Remembering | 6 |
| B) In the given network, K is closed at $t=0$ with zero current in the inductor. Find the values of i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t=0$, if $R=8\ \Omega$ and $L=0.2\text{ H}$. | Remembering | 6 |



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|---|---------------|---|
| C) In circuit shown in figure, the switch k is changed from position 1 to position 2 at $t=0$. A steady state condition is reached before switching. Find the values of i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t=0^+$. | Understanding | 6 |
|---|---------------|---|



Q.4 Solve Any Two of the following. 12

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|---|---------------|---|
| A) Z- Parameters of network are $Z_{11}=2\ \Omega$, $Z_{12}=1\ \Omega$, $Z_{21}=1\ \Omega$, $Z_{22}=3\ \Omega$. Find Y and ABCD parameters. | Understanding | 6 |
| B) Derive Z parameters in terms of Y parameters. | Understanding | 6 |
| C) What is the physical significance of Pole and Zero in a Transfer Function? | Remembering | 6 |

Q. 5 Solve Any Two of the following. 12

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|--|---------------|---|
| A) Explain High pass filter and band pass filter. | Understanding | 6 |
| B) What is meant by resonance in series RLC circuit? Derive equation for resonant frequency. | Remembering | 6 |
| C) Design a constant k LPF with $F_c = 1\text{ kHz}$ and $R_o = 600\ \Omega$. At what frequency attenuation (α) will be 10 dB. | Understanding | 6 |

*** End ***